

Pandas

```
!pip install pandas
```

```
Requirement already satisfied: pandas in /usr/local/lib/python3.12/dist-packages (2.2.2)  
Requirement already satisfied: numpy>=1.26.0 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.0.2)  
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas) (2.9.0.post0)  
Requirement already satisfied: pytz>=2020.1 in /usr/local/lib/python3.12/dist-packages (from pandas) (2025.2)  
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas) (2026.1)  
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas) (1.
```

```
import pandas as pd  
import numpy as np
```

```
print(pd.__version__)  
print(np.__version__)
```

```
2.2.2  
2.0.2
```

```
# creating List using Series  
a=pd.Series([1,2,3,4,5])  
print(a)
```

```
0    1  
1    2  
2    3  
3    4  
4    5  
dtype: int64
```

```
a=pd.Series([1,2,3,4,5],  
index=['a','b','c','d','e'])  
print(a)
```

```
a    1  
b    2  
c    3  
d    4  
e    5  
dtype: int64
```

```
student={  
    "Name":"VISHNU",  
    "department": "AIIML"  
}  
S=pd.Series(student)  
print(S)
```

```
Name          VISHNU  
department     AIIML  
dtype: object
```

```
m=pd.Series([20,30,40,50])  
print(m)
```

```
0    20  
1    30  
2    40  
3    50  
dtype: int64
```

```
m+2
```

```
      0  
-----  
0    22  
1    32  
2    42  
3    52  
  
dtype: int64
```

```
m/2
```

```
      0
0  10.0
1  15.0
2  20.0
3  25.0

dtype: float64
```

```
m-2
```

```
      0
0  18
1  28
2  38
3  48

dtype: int64
```

```
m%2
```

```
      0
0  0
1  0
2  0
3  0

dtype: int64
```

```
m.min()
```

```
20
```

```
m.max()
```

```
50
```

```
m.describe()
```

```
      0
count  4.000000
mean   35.000000
std    12.909944
min    20.000000
25%    27.500000
50%    35.000000
75%    42.500000
max    50.000000

dtype: float64
```

```
m.sum()
```

```
np.int64(140)
```

```
m[m>15]
```

```
0
0 20
1 30
2 40
3 50

dtype: int64
```

```
name=pd.Series(["Vishnu","beja","Hello"])
name.str.lower()
```

```
0
0 vishnu
1 beja
2 hello

dtype: object
```

```
name.str.len()
```

```
0
0 6
1 4
2 5

dtype: int64
```

```
name.str.upper()
```

```
0
0 VISHNU
1 BEJA
2 HELLO

dtype: object
```

```
name.str.startswith("V")
```

```
0
0 True
1 False
2 False

dtype: bool
```

```
name.str.endswith("a")
```

```
0
0 False
1 True
2 False

dtype: bool
```

Data frame

```
student={
    "names":["Vishnu","Beja","Sathish","Manoj"],
    "age":[22,21,20,21],
    "department":["AIML","AIML","AIML","AIML"]
}
```

```
a=pd.DataFrame(student)
print(a)
```

	names	age	department
0	Vishnu	22	AIML
1	Beja	21	AIML
2	Sathish	20	AIML
3	Manoj	21	AIML

```
Rows=[(
    "Vishnu",22,"AIML"
),
(
    "Beja",21,"AIML"
),
(
    "Sathish",20,"AIML"
),
(
    "Manoj",21
)]
ds=pd.DataFrame(Rows)
print(ds)
```

	0	1	2
0	Vishnu	22	AIML
1	Beja	21	AIML
2	Sathish	20	AIML
3	Manoj	21	None

```
#from numpy array
arr=np.random.rand(3,4)
c=pd.DataFrame(arr,columns=['A','B','C','D'],index=['a','b','c'])
print(c)
```

	A	B	C	D
a	0.000683	0.431819	0.999741	0.812475
b	0.689699	0.904884	0.463643	0.420200
c	0.586667	0.459056	0.132246	0.876141

```
#Exploring Data Frames
data ={
    'Name':['Vishnu','Beja','Sathish','Manoj'],
    'Age':[22,21,20,21],
    'Score':['99','95','90','89'],
    'Grade':['A','A','A','B']
}
df=pd.DataFrame(data)
print(df)
```

	Name	Age	Score	Grade
0	Vishnu	22	99	A
1	Beja	21	95	A
2	Sathish	20	90	A
3	Manoj	21	89	B

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 4 entries, 0 to 3
Data columns (total 4 columns):
#   Column  Non-Null Count  Dtype
---  -
0    Name    4 non-null        object
1    Age      4 non-null        int64
2    Score    4 non-null        object
3    Grade    4 non-null        object
dtypes: int64(1), object(3)
memory usage: 260.0+ bytes
```

```
df.describe()
```

```
      Age
count  4.000000
mean   21.000000
std     0.816497
min    20.000000
25%    20.750000
50%    21.000000
75%    21.250000
max    22.000000
```

```
df.head()
```

```
      Name  Age  Score  Grade
0  Vishnu   22     99      A
1   Beja   21     95      A
2 Sathish   20     90      A
3  Manoj   21     89      B
```

Next steps:

[Generate code with df](#)[New interactive sheet](#)

```
df.index
```

```
RangeIndex(start=0, stop=4, step=1)
```

```
df.columns
```

```
Index(['Name', 'Age', 'Score', 'Grade'], dtype='object')
```

```
df.dtypes
```

```
      0
Name  object
Age    int64
Score  object
Grade  object

dtype: object
```

```
df.nunique()
```

```
      0
Name   4
Age    3
Score  4
Grade  2

dtype: int64
```

```
df.isnull().sum()
```

```
      0
Name   0
Age    0
Score  0
Grade  0

dtype: int64
```

```
df.tail()
```

	Name	Age	Score	Grade
0	Vishnu	22	99	A
1	Beja	21	95	A
2	Sathish	20	90	A
3	Manoj	21	89	B

Selecting ,Filtering Data

```
#Selecting Columns
df['Name'] #Single column
df.Name # or any 1 cmd
```

	Name
0	Vishnu
1	Beja
2	Sathish
3	Manoj

dtype: object

```
#Multiple columns
df[['Name','Age']]
```

	Name	Age
0	Vishnu	22
1	Beja	21
2	Sathish	20
3	Manoj	21

loc

```
df.loc[0]
```

	0
Name	Vishnu
Age	22
Score	99
Grade	A

dtype: object

```
df.loc[0:1]
```

	Name	Age	Score	Grade
0	Vishnu	22	99	A
1	Beja	21	95	A

```
df.loc[0:2, 'Name':'Score']
```

	Name	Age	Score
0	Vishnu	22	99
1	Beja	21	95
2	Sathish	20	90

```
df.iloc[0]
```

```
0
Name Vishnu
Age 22
Score 99
Grade A
dtype: object
```

```
df.iloc[-1]
```

```
3
Name Manoj
Age 21
Score 89
Grade B
dtype: object
```

```
df.iloc[0:2]
```

```
   Name  Age  Score  Grade
0  Vishnu  22    99     A
1   Beja  21    95     A
```

```
df.iloc[0:2,0:2]
```

```
   Name  Age
0  Vishnu  22
1   Beja  21
```

```
#Single Value
df.loc[0, 'Name']
```

```
'Vishnu'
```

```
df.iloc[0,0]
```

```
'Vishnu'
```

```
Start coding or generate with AI.
```

```
Start coding or generate with AI.
```